Università degli Studi di Milano

Tesi di laurea magistrale
Scienze e Tecnologie: Informatica

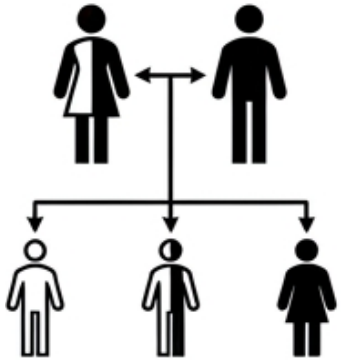
Patient Knowledge Graph Representation using Non-Euclidean metric

Andrea Moretti

Relatore: Mauricio Soto Gomez
Correlatrice: Elena Casiraghi

La problematica affrontata: malattie mendeliane

Mutazione monogenetica



Ereditarietà

Secondo il National Institutes of Health (NIH)

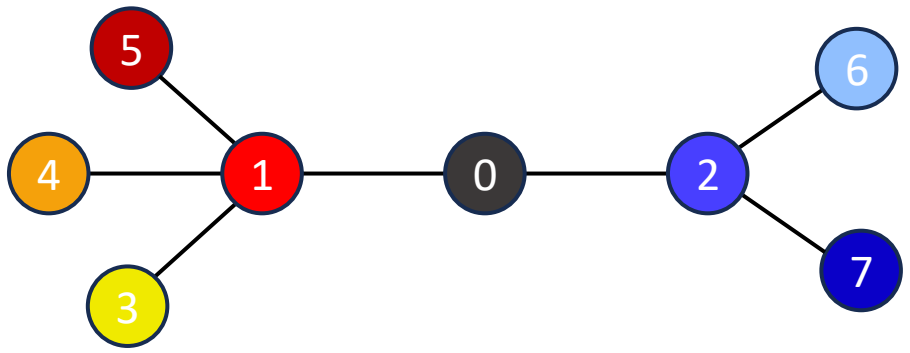
Esistono 10 000 malattie mendeliane distinte

Diagnosticate circa 2 milioni di persone l'anno

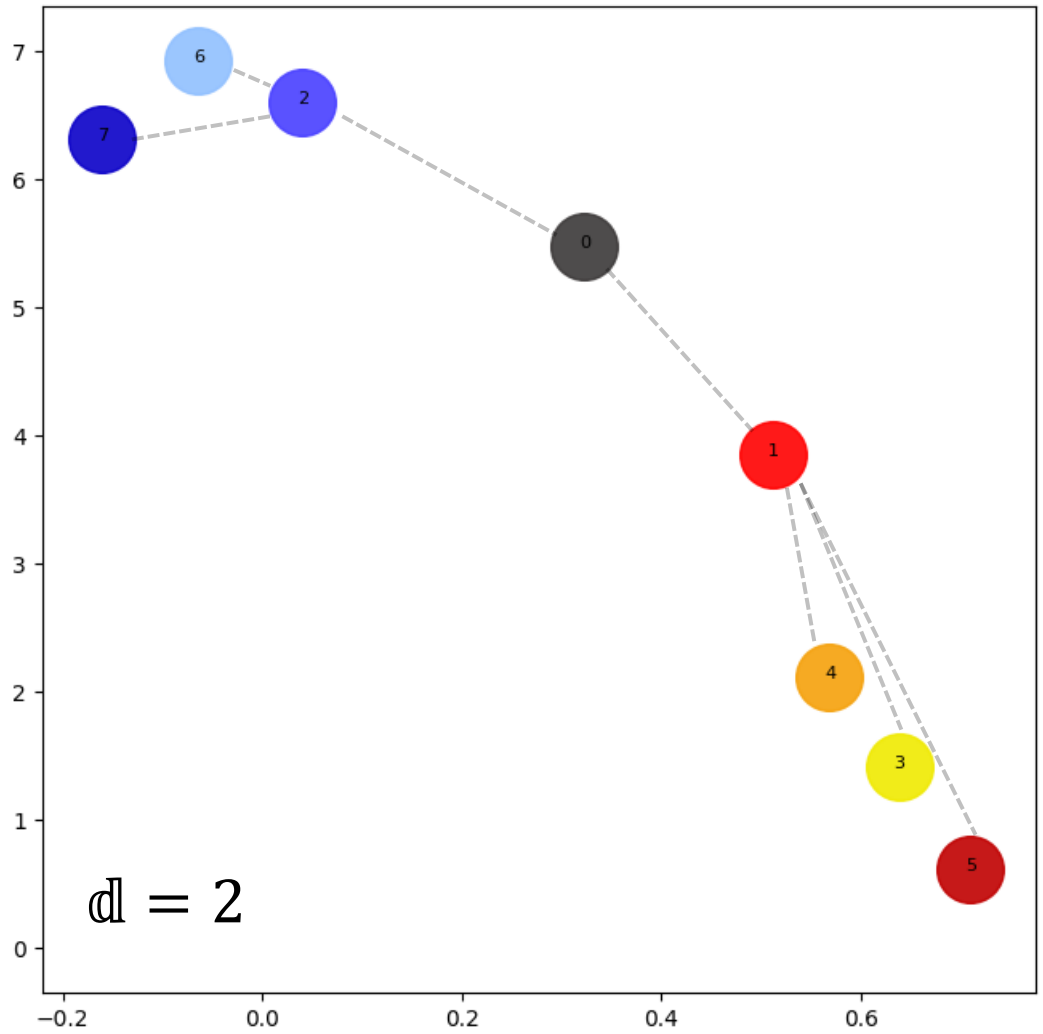
Fonte: American Society of Human Genetics **[1]**

Oltre il 50% dei pazienti non riceve una diagnosi molecolare certa

Il graph embedding



f_{map}



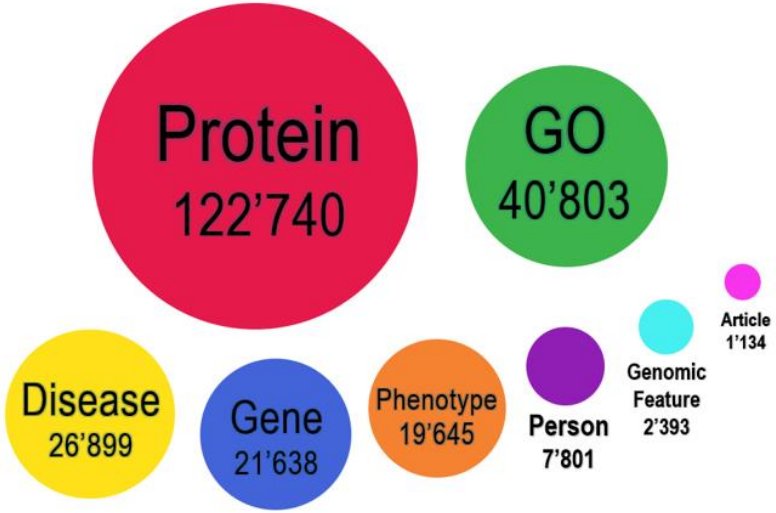
Formalmente: $f_{\text{map}}: V \rightarrow \mathbb{R}^{\mathbb{d}}$

Obiettivi:

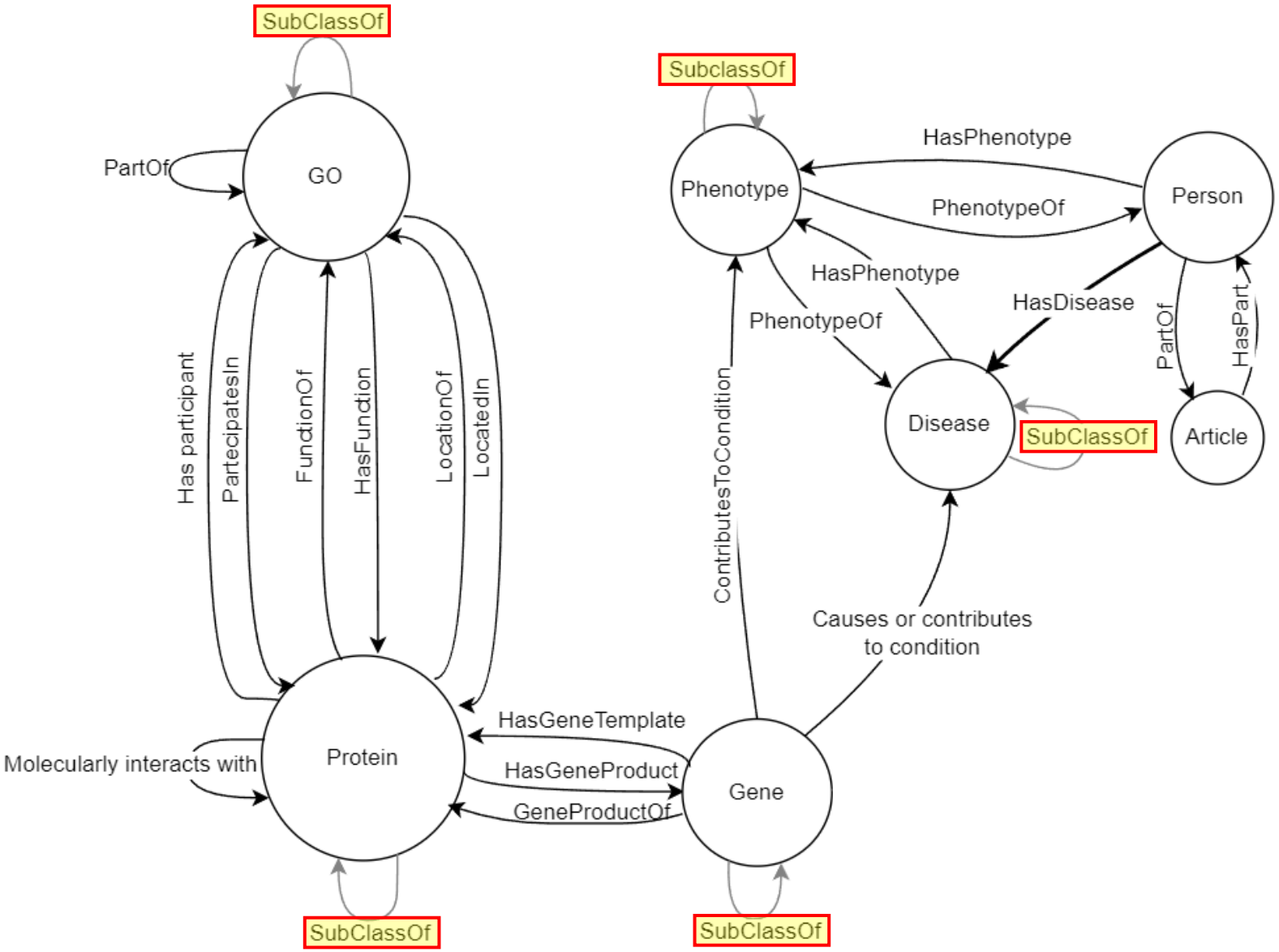
- Preservare la topologia
- Ridurre la dimensionalità

Caso studio

Proporzioni del numero di nodi presenti

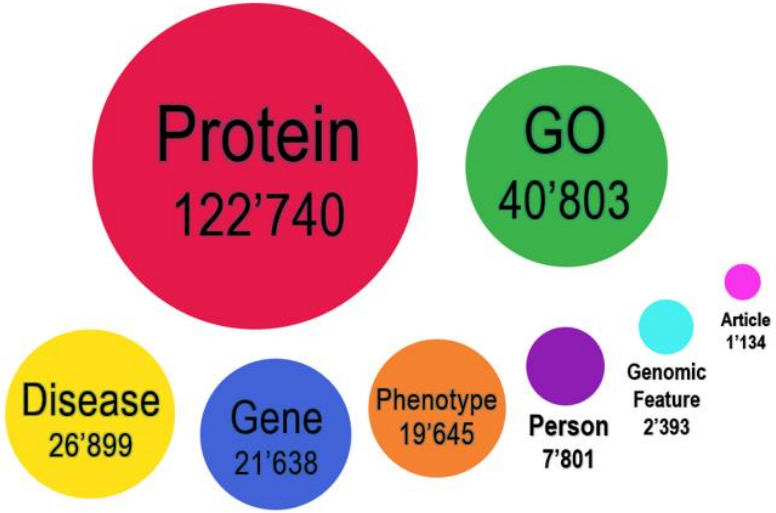


# Nodi	# Archi
243 053	2 723 540



Caso studio

Proporzioni del numero di nodi presenti



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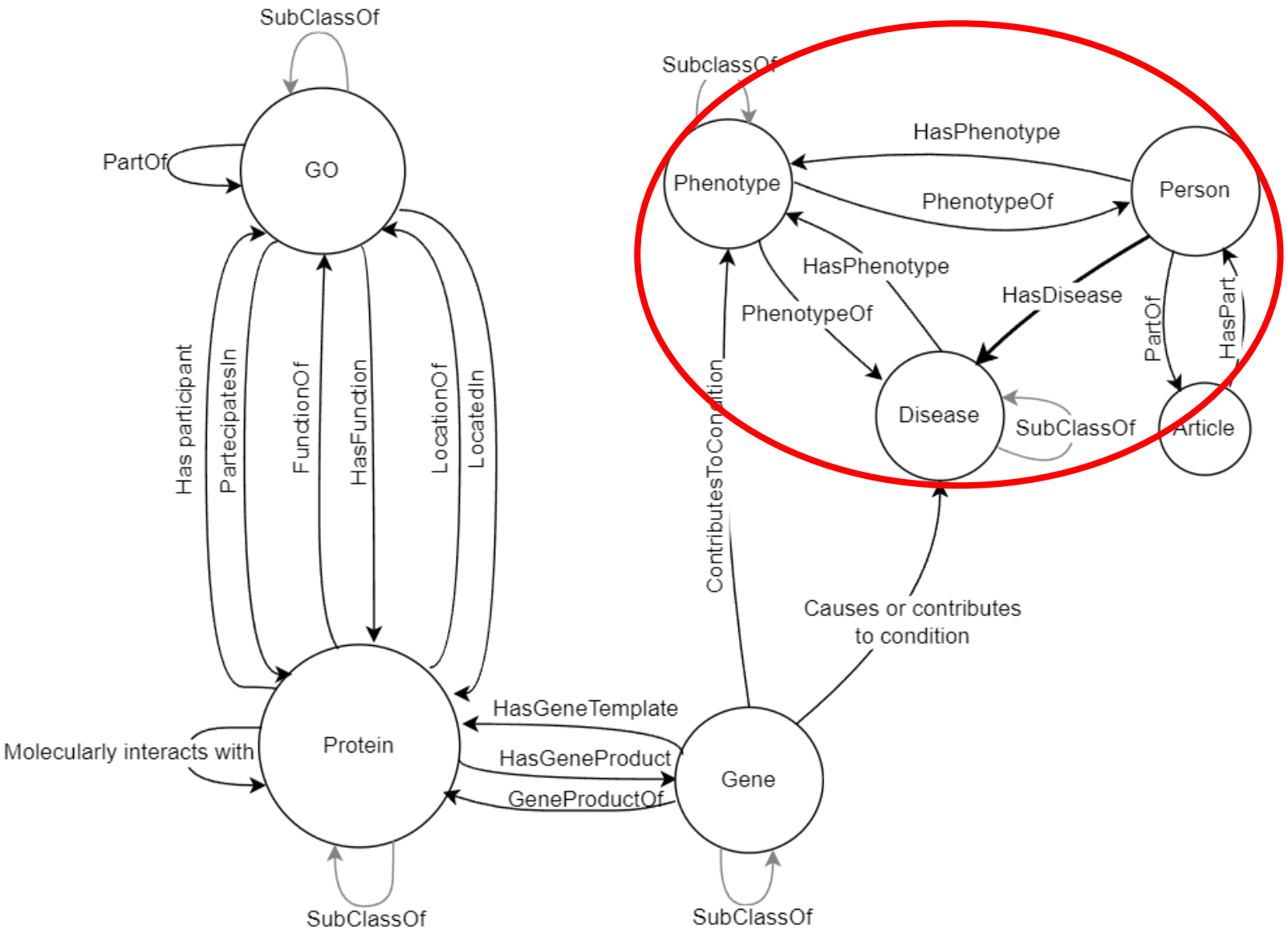


Caso studio

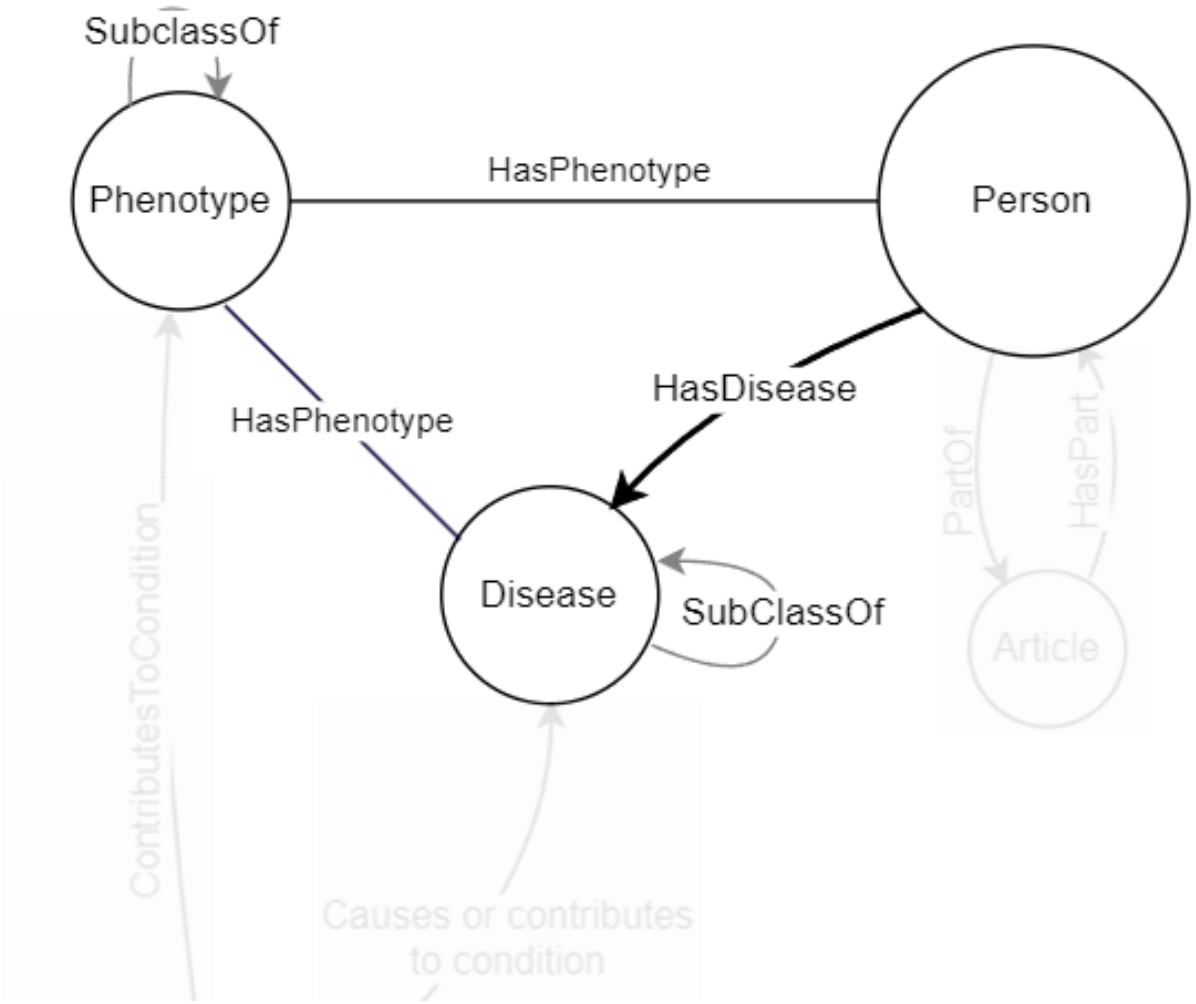
Prima problematica riscontrata:

– Inefficienza della libreria nella gestione della memoria

# Nodi	# Archi
243 053	2 723 540

↓ ↓

# Nodi	# Archi
28 811	178 320



Obiettivi della ricerca

Primo studio

Scelta dello spazio per il node embedding

Secondo studio

Studio della tipologia di modelli (omogeneo o eterogeneo)

Terzo studio

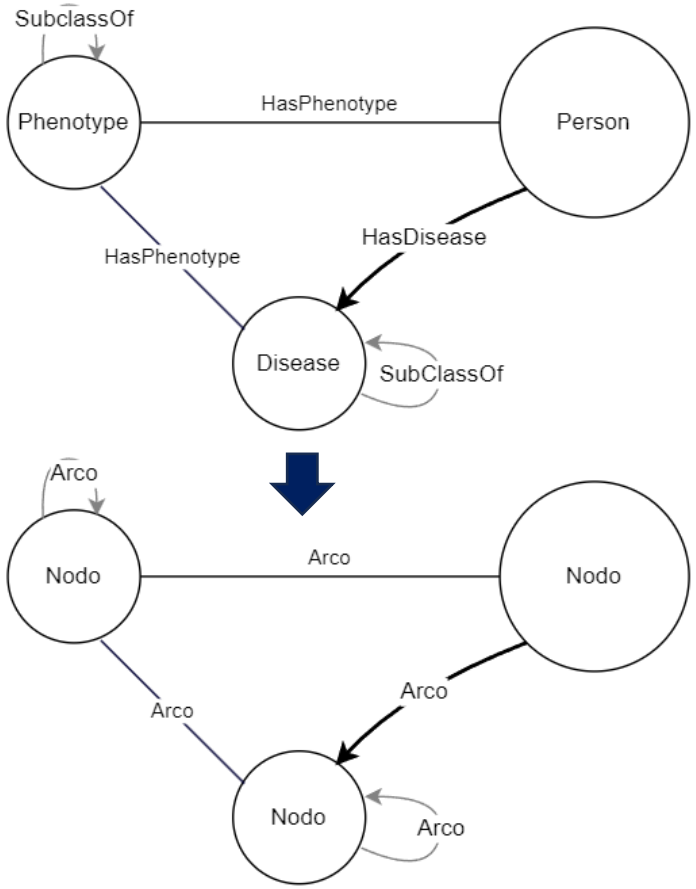
Analisi della coerenza dei dati biomedici

Obiettivo finale

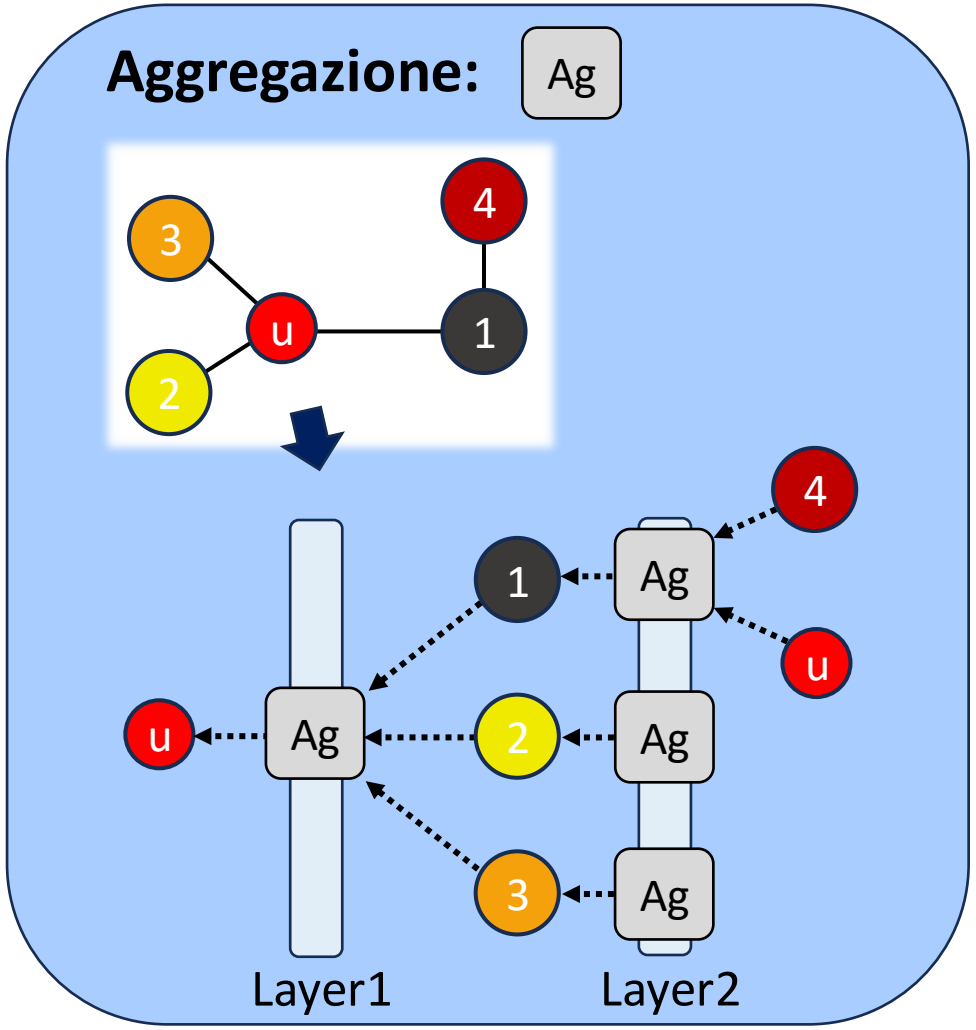
Supporto alla diagnosi di malattie mendeliane tramite Knowledge Graph Embedding

Modelli omogenei: GCN e HGCN [2]

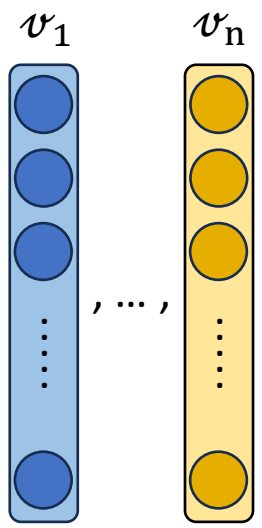
Grafo in input:



Aggregazione:

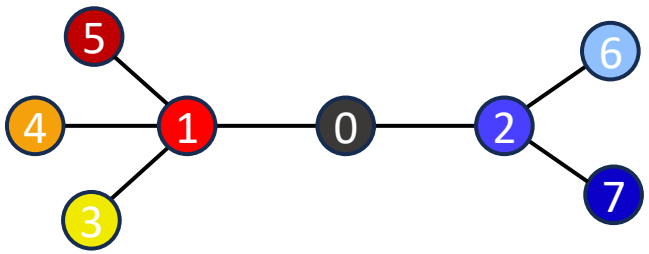


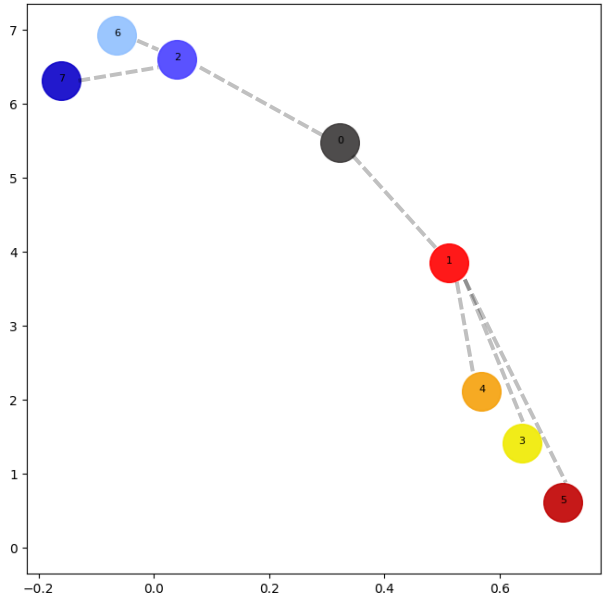
Embedding:



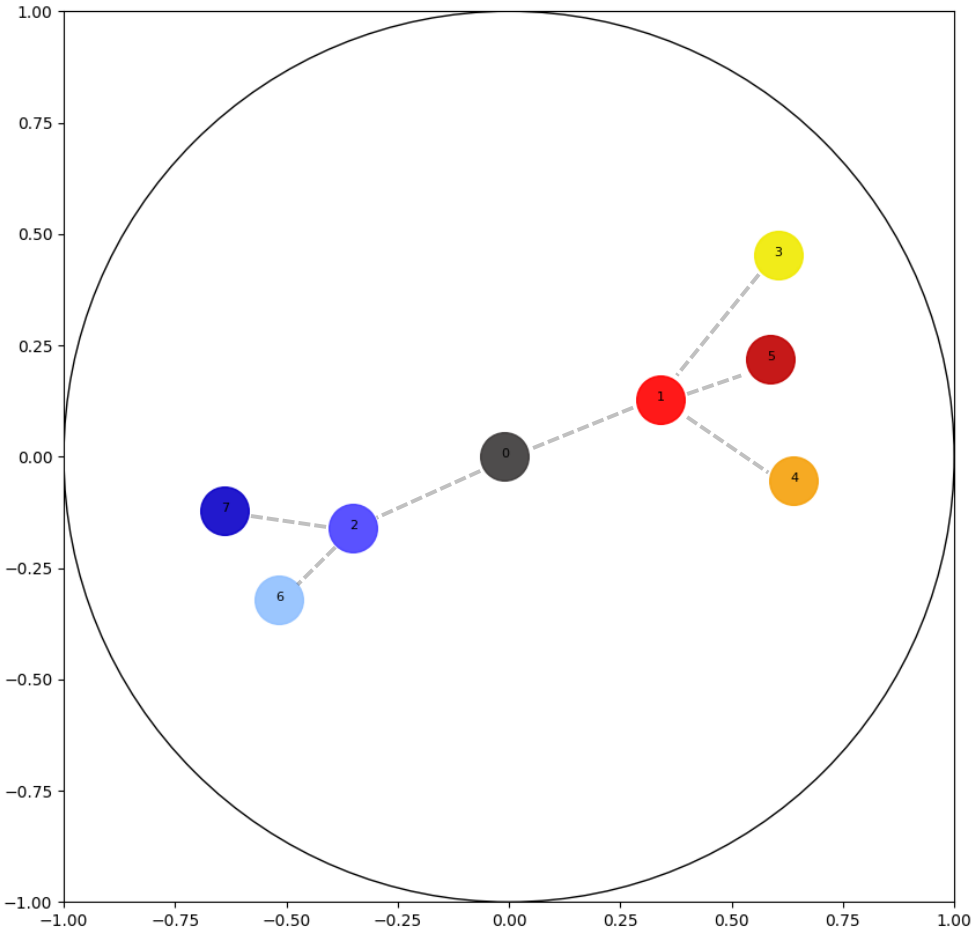
Modelli omogenei: GCN e HGCN [2]

Euclideo:

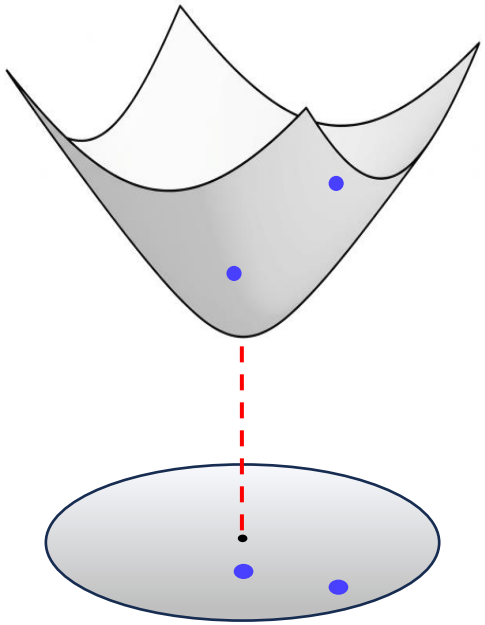




Iperbolico:



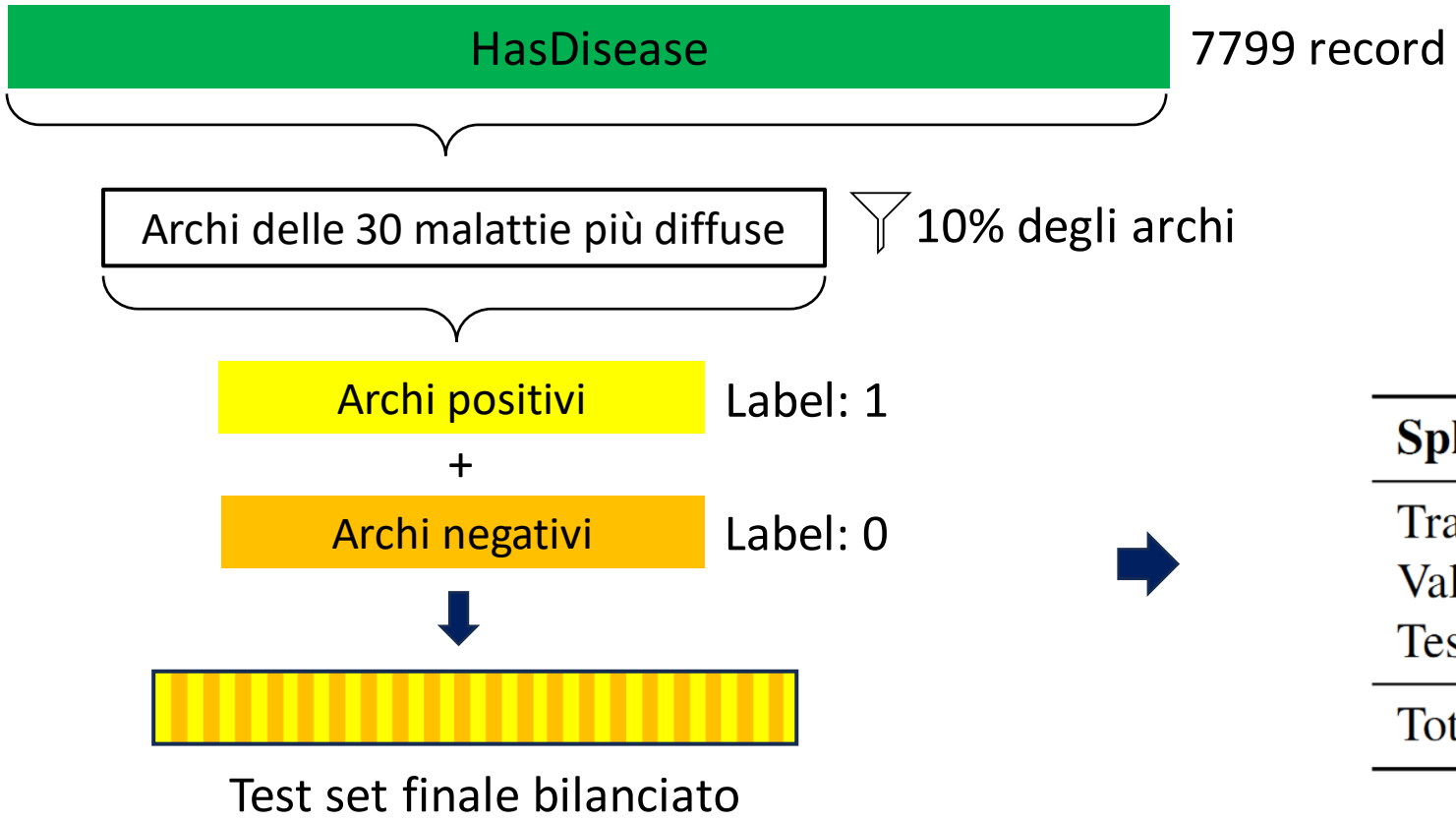
Disco di Poincaré

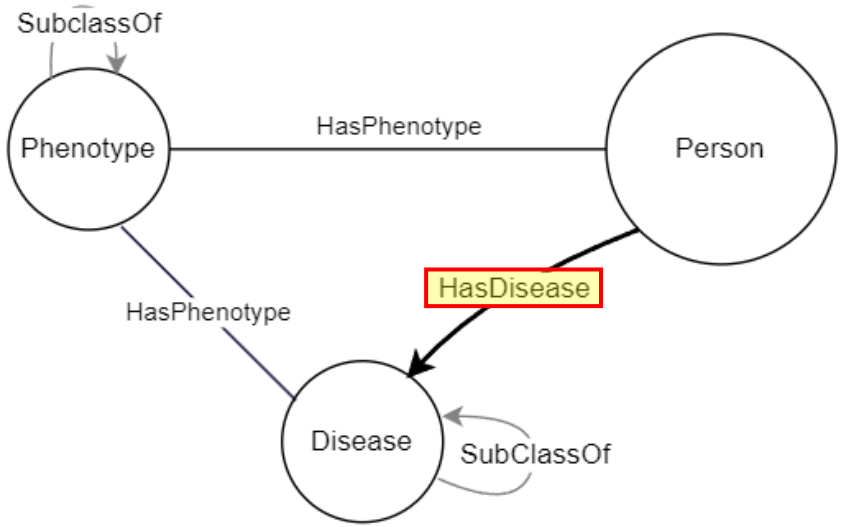


Esempio con $\mathbb{d} = 2$

Configurazione sperimentale

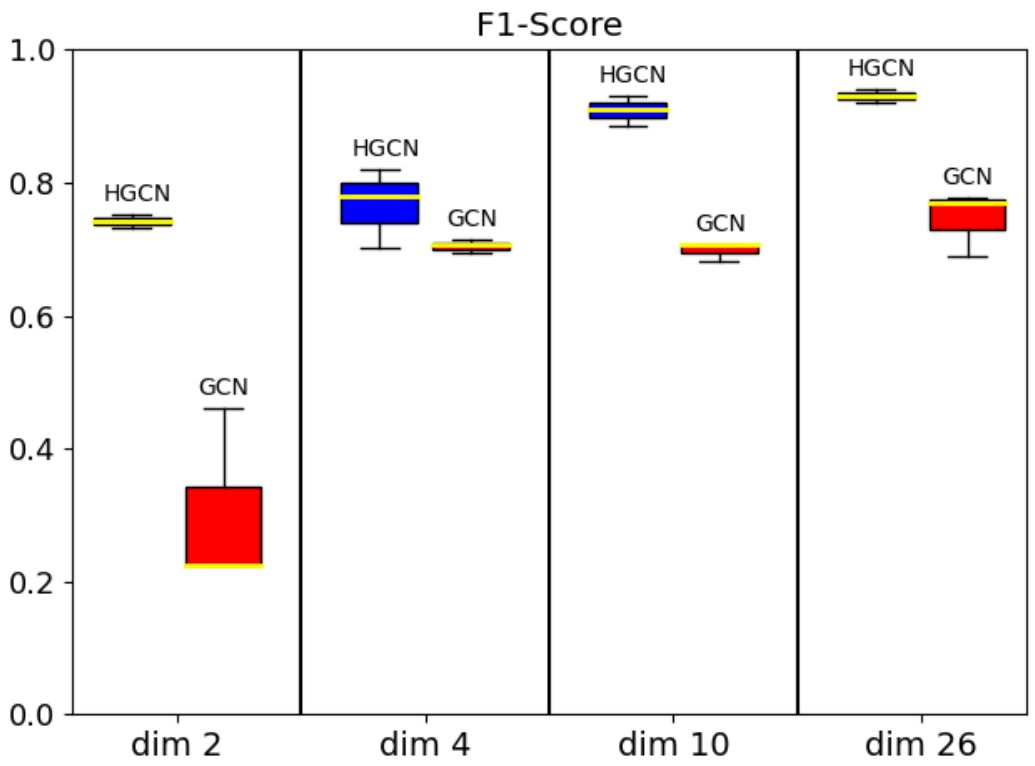
Obiettivo: predire associazione tra paziente e malattia





Split	Numero di archi	%
Train	168 624	94.5
Validation	8 136	4.5
Test	1 560	1.0
Totale	178 320	100

Applicazione GCN e HGCN al caso studio



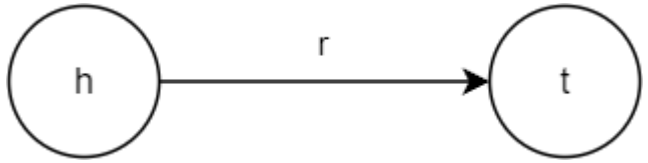
Performance medie su 3 seed differenti

Metric	dim 2		dim 4		dim 10		dim 26	
	GCN	HGCN	GCN	HGCN	GCN	HGCN	GCN	HGCN
ROC-AUC	0.55	**0.74**	0.74	**0.83**	0.67	**0.96**	0.73	**0.97**
Precision	0.24	**0.66**	0.58	**0.76**	0.58	**0.91**	0.62	**0.94**
Recall	0.42	**0.85**	**0.91**	0.79	0.90	**0.91**	**0.94**	0.92
F1-score	0.30	**0.74**	0.70	**0.77**	0.70	**0.91**	0.75	**0.93**

I modelli eterogenei: AttE e AttH [3]

Knowledge Graph Embedding:

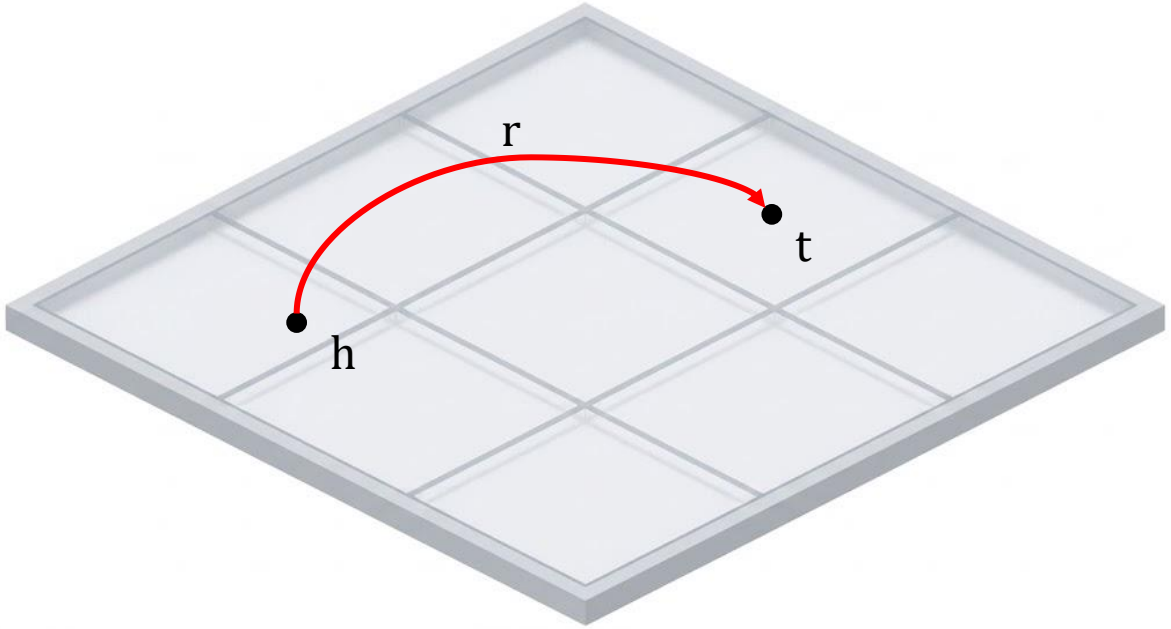
Impara vettori distinti per entità (h, t) e per relazioni (r)



Attention:

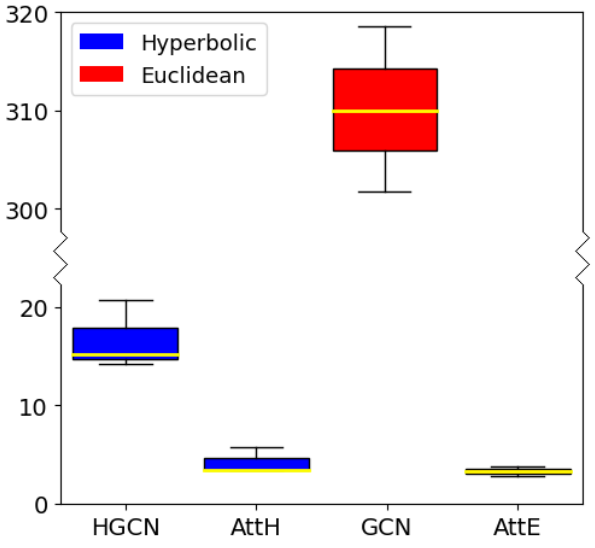
Assegna pesi dinamici per valorizzare le associazioni cliniche più rilevanti

Formalmente: $h \circ r \approx t$

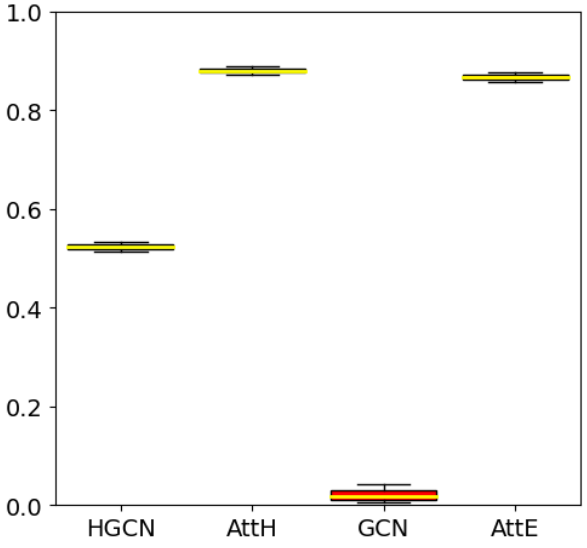


Valutazione efficienza delle predizioni (dim 26)

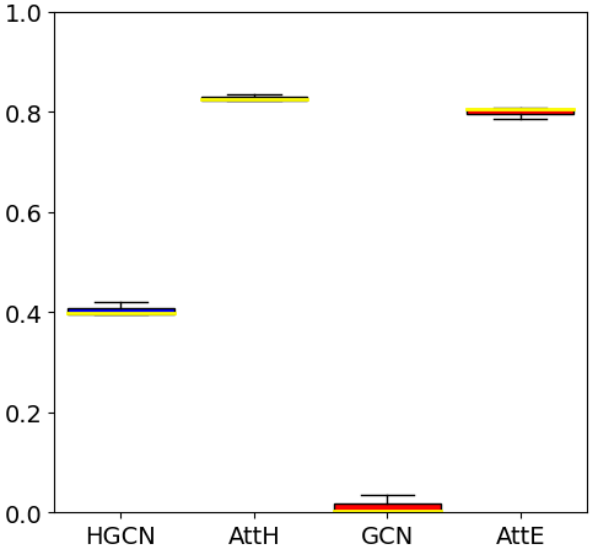
Mean rank (MR) ↓



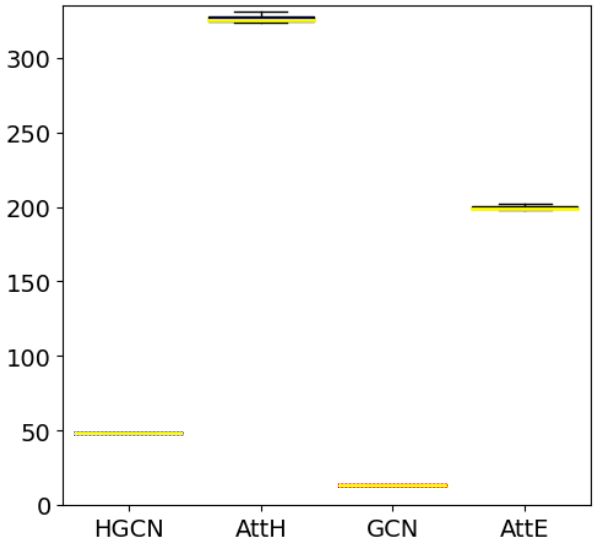
Mean reciprocal rank (MRR)



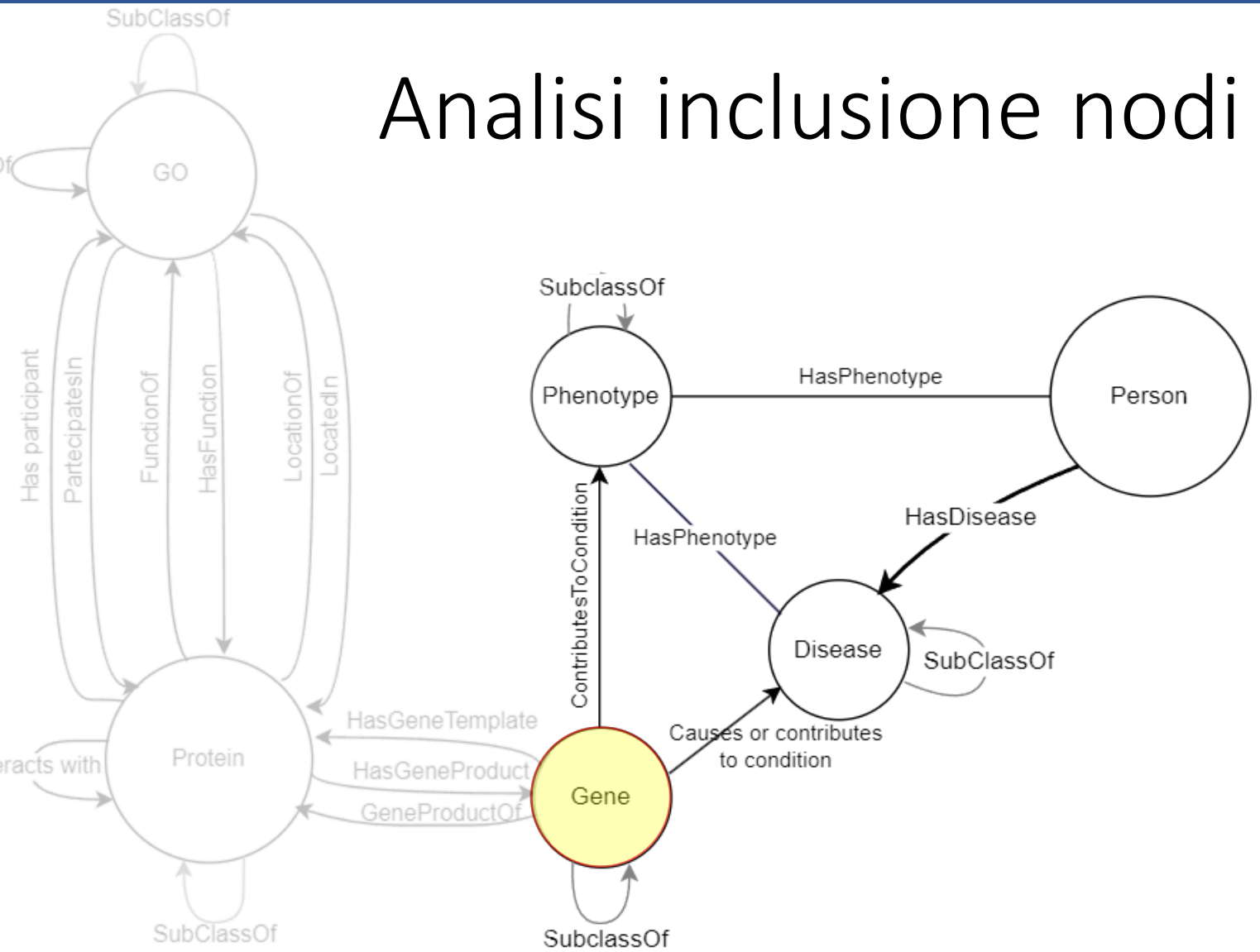
Hits@1



Tempo totale (sec)



Analisi inclusione nodi di tipo Gene



# Nodi	# Archi
28 811	178 320

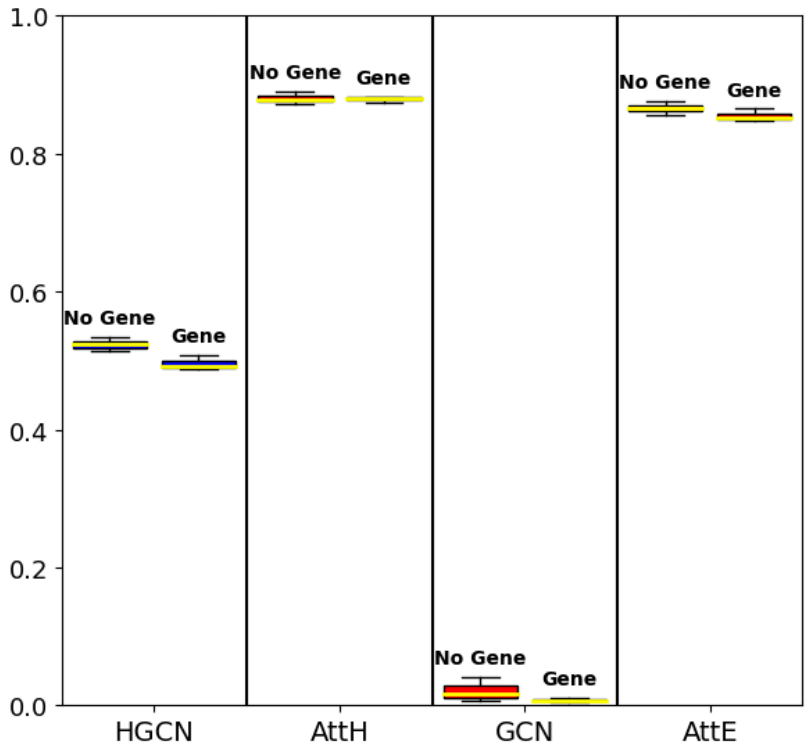
↓ ↓

# Nodi	# Archi
36 171	208 338

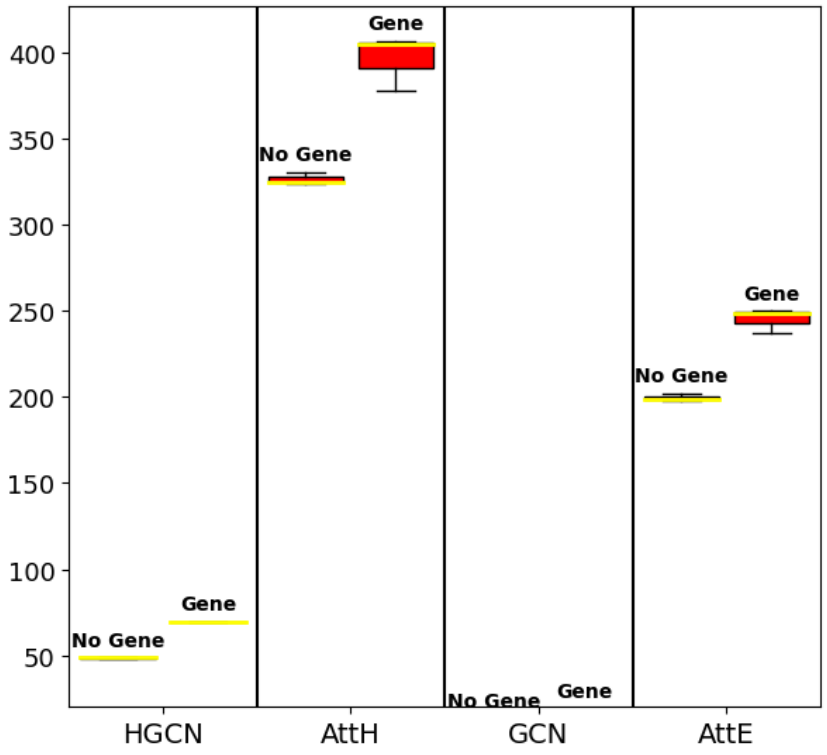
Split	Numero di archi	%
Train	197 142	94.6
Validation	9 636	4.6
Test	1 560	0.8
Totale	208 338	100

Analisi inclusione nodi di tipo Gene (dim 26)

Mean reciprocal rank (MRR)

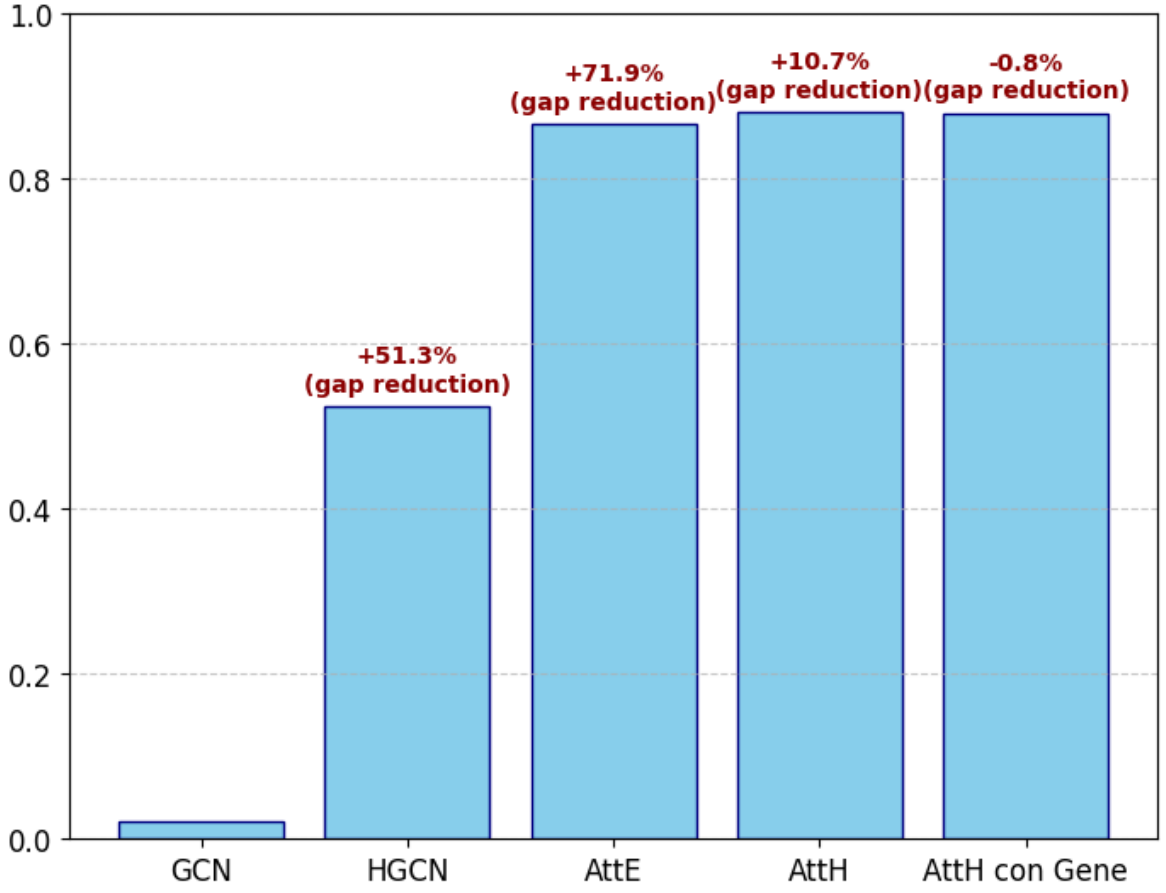


Tempo totale (sec)

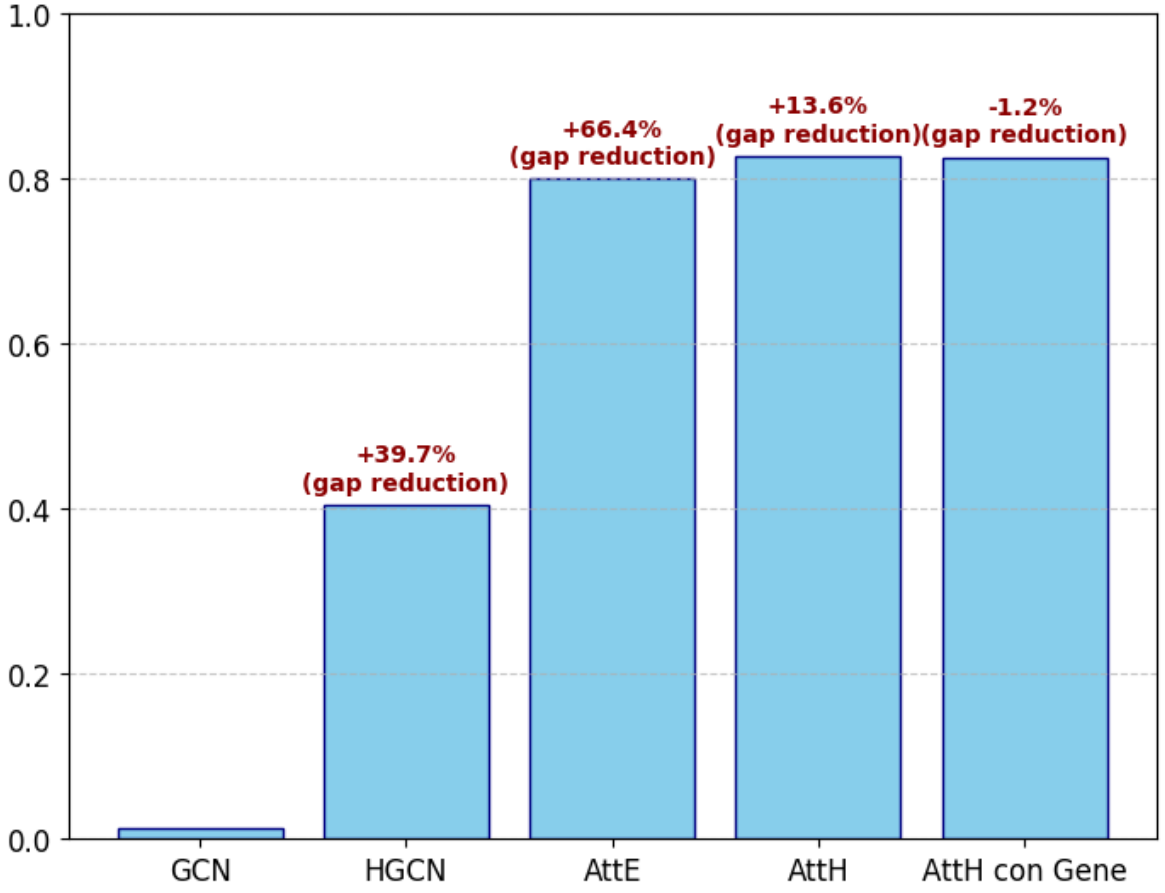


Confronto finale

Mean reciprocal rank (MRR)



Hits@1



Grazie per l’attenzione

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Fonti principali utilizzate

[1]: Monica H. Wojcik, et al.
Beyond the exome: What's next in diagnostic testing for Mendelian conditions. (2023)
In: The American Journal of Human Genetics

[2]: Ines Chami, et al.
Hyperbolic Graph Convolutional Neural Networks. (2019).
https://github.com/HazyResearch/hgcn.

[3]: Ines Chami, et al.
Low-Dimensional Hyperbolic Knowledge Graph Embeddings. (2020)
https://github.com/HazyResearch/KGEmb.